

Power Supply Systems

Part II

Phase 5: Distributor Fed at Both Ends

Aim: To learn distributor fed at both ends from absolute zero and become ready for semester numericals.

Style: Simple English, SN ma'am style, clean diagrams, no overlapping labels, every diagram word explained.

Target: Equal end voltages, unequal end voltages, current supplied from both ends, minimum potential point, voltage calculation, uniformly distributed load and power loss.

Subject: Power Supply Systems

Section: Part II

Phase: 5 – Distributor Fed at Both Ends

Level: Absolute zero to semester-question-ready

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1 Phase 5 Roadmap

Phase 5 Goal

In Phase 4, we studied a distributor fed from one end only. Now we study a better arrangement where the distributor receives supply from both ends. This improves voltage regulation and reliability.

Supply at A and B → Loads between A and B → Current from both ends → Minimum potential point → Voltage

Memory Hook

In one-end feeding, all current comes from one side. In both-end feeding, some current comes from A and the remaining current comes from B.

2 Meaning of Distributor Fed at Both Ends

Definition

A **distributor fed at both ends** is a distributor in which supply is given from both ends of the distributor.

The two end voltages may be:

1. equal end voltages,

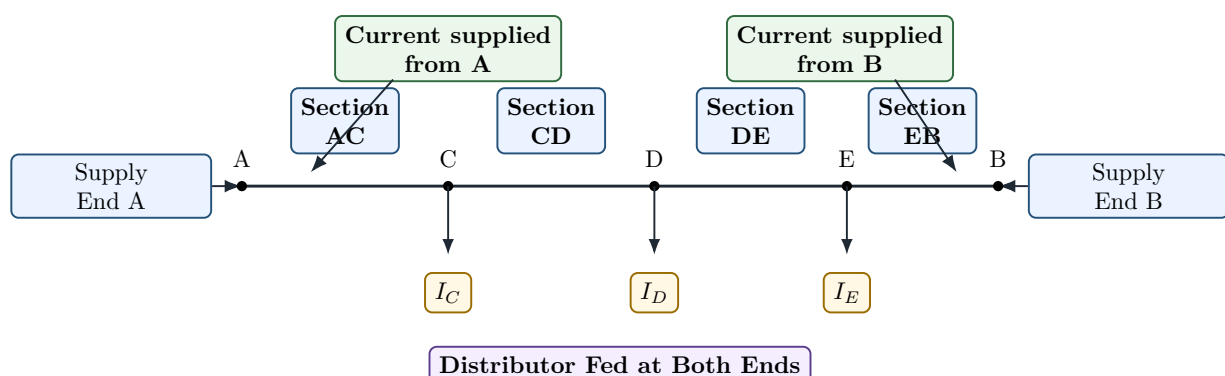
$$V_A = V_B$$

2. unequal end voltages,

$$V_A \neq V_B$$

Both-end feeding gives better voltage regulation than one-end feeding.

3 Basic Diagram of Distributor Fed at Both Ends



How to Read the Diagram

Word/Symbol	Meaning
Supply End A	Left side supply point of the distributor. It has voltage V_A .
Supply End B	Right side supply point of the distributor. It has voltage V_B .
A and B	The two ends of the distributor. Supply is given from both these points.
C, D, E	Load tapping points between A and B.
I_C, I_D, I_E	Load currents taken by consumers at C, D and E.
Section AC, CD, DE, EB	Different conductor sections of the distributor. Each section has its own current and resistance.
Current supplied from A	Part of the total load current entering from left end A.
Current supplied from B	Remaining part of the total load current entering from right end B.
Distributor Fed at Both Ends	The complete arrangement where loads are supplied from both A and B.

4 Why Both-End Feeding is Better

Compared to one-end feeding:

1. Voltage drop is less.
2. Minimum voltage improves.
3. Supply is more reliable.
4. Current is shared between two ends.
5. If one end has a fault or is disconnected, supply may be maintained from the other end.

Semester Answer Style

Semester line:

Both-end feeding gives better voltage regulation because current is shared by the two feeding ends.

5 Minimum Potential Point

Definition

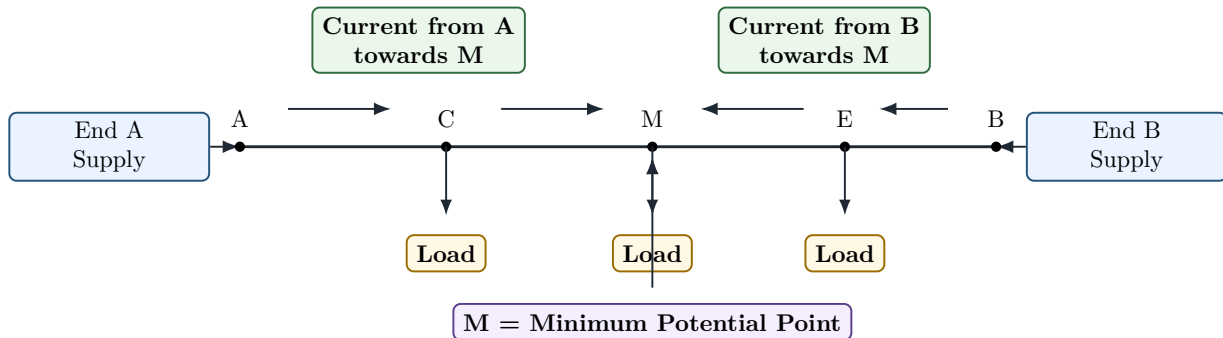
The **minimum potential point** is the point on the distributor where voltage is minimum.

In a distributor fed at both ends:

- current comes from A on the left side,
- current comes from B on the right side,
- at one point, the direction of current changes,
- that point is the point of minimum potential.

At minimum potential point, current changes direction.

6 Diagram of Minimum Potential Point



How to Read the Diagram

Word/Symbol	Meaning
End A Supply	Left end supply. It sends current towards the loads on left part.
End B Supply	Right end supply. It sends current towards the loads on right part.
M	Minimum potential point. Voltage is lowest at this point.
Current from A towards M	Current enters from A and flows up to the minimum potential point.
Current from B towards M	Current enters from B and flows up to the minimum potential point.
Current direction arrows	They show that current approaches point M from both sides.
Load	Consumer current tapped from the distributor.
Minimum Potential Point	The point where current changes direction and voltage becomes minimum.

7 Case 1: Both Ends Fed at Equal Voltage

For equal end voltages:

$$V_A = V_B$$

This means the total voltage drop from A to B must be zero when calculated algebraically.

$$\sum IR = 0$$

This is the KVL idea used in both-end distributor problems.

8 Current Supplied from Each End: Moment Method

Consider a distributor AB of total length L . Loads are connected at distances $x_1, x_2, x_3, \dots$ from end A.

Let:

I_A = current supplied from A

I_B = current supplied from B

For equal end voltages:

$$I_A = \frac{\sum(\text{load current} \times \text{distance from B})}{L}$$

$$I_B = \frac{\sum(\text{load current} \times \text{distance from A})}{L}$$

Also,

$$I_A + I_B = \text{total load current}$$

Memory Hook

For equal end voltages:

Current from A uses distance from B

Current from B uses distance from A

9 Derivation of Moment Method

Let total length be L . Let loads $I_1, I_2, I_3, \dots$ be placed at distances $x_1, x_2, x_3, \dots$ from A.

Then distance of load I_1 from B is:

$$L - x_1$$

distance of load I_2 from B is:

$$L - x_2$$

and so on.

Let resistance per metre be r . For equal end voltages:

$$V_A = V_B$$

So algebraic drop from A to B is zero.

$$r [I_A L - I_1(L - x_1) - I_2(L - x_2) - I_3(L - x_3) - \dots] = 0$$

Since $r \neq 0$,

$$I_A L = I_1(L - x_1) + I_2(L - x_2) + I_3(L - x_3) + \dots$$

Therefore,

$$I_A = \frac{I_1(L - x_1) + I_2(L - x_2) + I_3(L - x_3) + \dots}{L}$$

Similarly,

$$I_B = \frac{I_1x_1 + I_2x_2 + I_3x_3 + \dots}{L}$$

10 Solved Example 1: Equal End Voltages

10.1 Problem

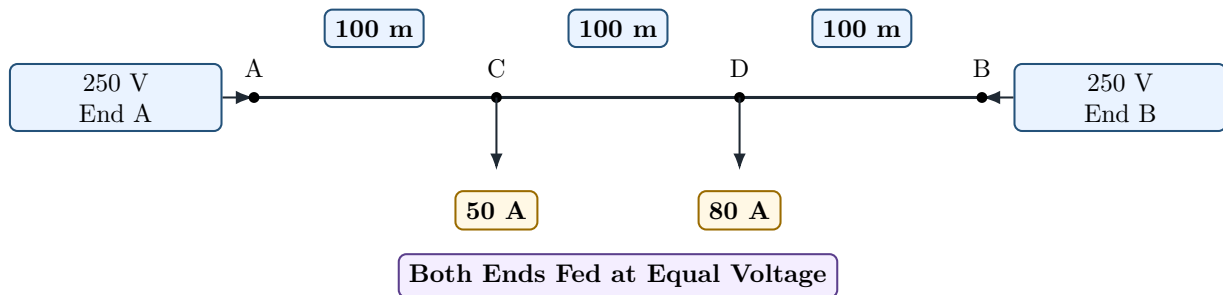
A 2-wire DC distributor AB is 300 m long and is fed from both ends at 250 V. Two loads are connected:

50 A at 100 m from A

80 A at 200 m from A

Resistance of the distributor is $0.0005\Omega/\text{m}$ for both conductors. Find the currents supplied from A and B and the minimum voltage.

10.2 Line Diagram



How to Read the Diagram

Word/Symbol	Meaning
250 V End A	Left supply end. Voltage at A is 250 V.
250 V End B	Right supply end. Voltage at B is also 250 V.
A and B	Two feeding ends of the distributor.
C	Load point 100 m from A. Load at C is 50 A.
D	Load point 200 m from A. Load at D is 80 A.
100 m	Length of each section AC, CD and DB.
Both Ends Fed at Equal Voltage	Both end voltages are the same, so $V_A = V_B = 250 \text{ V}$.

10.3 Step 1: Current Supplied from A

Total length:

$$L = 300 \text{ m}$$

For 50 A load, distance from B is:

$$300 - 100 = 200 \text{ m}$$

For 80 A load, distance from B is:

$$300 - 200 = 100 \text{ m}$$

$$I_A = \frac{50 \times 200 + 80 \times 100}{300}$$

$$I_A = \frac{10000 + 8000}{300}$$

$$\boxed{I_A = 60 \text{ A}}$$

10.4 Step 2: Current Supplied from B

Total load:

$$I_{\text{total}} = 50 + 80 = 130 \text{ A}$$

$$I_B = 130 - 60$$

$$\boxed{I_B = 70 \text{ A}}$$

10.5 Step 3: Section Currents

From A to C:

$$I_{AC} = 60 \text{ A}$$

After 50 A load at C:

$$I_{CD} = 60 - 50 = 10 \text{ A}$$

After 80 A load at D:

$$I_{DB} = 10 - 80 = -70 \text{ A}$$

The negative sign means actual current in DB flows from B to D.

10.6 Step 4: Voltage Calculation

Resistance of each 100 m section:

$$R = 100 \times 0.0005 = 0.05\Omega$$

Voltage drop from A to C:

$$\Delta V_{AC} = 60 \times 0.05 = 3 \text{ V}$$

$$V_C = 250 - 3 = 247 \text{ V}$$

Voltage drop from C to D:

$$\Delta V_{CD} = 10 \times 0.05 = 0.5 \text{ V}$$

$$V_D = 247 - 0.5 = 246.5 \text{ V}$$

From B side:

$$\Delta V_{BD} = 70 \times 0.05 = 3.5 \text{ V}$$

$$V_D = 250 - 3.5 = 246.5 \text{ V}$$

So the calculation is correct.

$V_{\min} = 246.5 \text{ V at point D}$

Semester Answer Style

The minimum potential point is D because the current changes direction at D.

11 Case 2: Both Ends Fed at Unequal Voltages

For unequal end voltages:

$V_A \neq V_B$

Now the end with higher voltage will push extra current into the distributor.

If $V_A > V_B$, then current supplied from A increases.

If $V_B > V_A$, then current supplied from B increases.

12 Formula for Unequal End Voltages

Let:

$$V_A = \text{voltage at A}$$

V_B = voltage at B

L = total length of distributor

r = resistance per metre

I_A = current supplied from A

For loads $I_1, I_2, I_3, \dots$ placed at distances $x_1, x_2, x_3, \dots$ from A:

$$V_A - V_B = r [I_A L - I_1(L - x_1) - I_2(L - x_2) - I_3(L - x_3) - \dots]$$

Therefore,

$$I_A = \frac{V_A - V_B}{rL} + \frac{I_1(L - x_1) + I_2(L - x_2) + I_3(L - x_3) + \dots}{L}$$

Then,

$$I_B = I_{\text{total}} - I_A$$

Common Mistake

In unequal-voltage problems, do not use only the simple moment method. You must include the voltage difference term:

$$\frac{V_A - V_B}{rL}$$

13 Solved Example 2: Unequal End Voltages

13.1 Problem

A 2-wire DC distributor AB is 300 m long. End A is maintained at 250 V and end B is maintained at 245 V. Loads are:

50 A at 100 m from A

80 A at 200 m from A

Resistance of distributor is $0.0005\Omega/\text{m}$ for both conductors. Find currents supplied from A and B and the minimum voltage.

13.2 Step 1: Given Data

$$V_A = 250 \text{ V}$$

$$V_B = 245 \text{ V}$$

$$V_A - V_B = 5 \text{ V}$$

$$L = 300 \text{ m}$$

$$r = 0.0005 \Omega/\text{m}$$

13.3 Step 2: Equal-Voltage Current from A

From moment method:

$$I_{A0} = \frac{50(300 - 100) + 80(300 - 200)}{300}$$

$$I_{A0} = 60 \text{ A}$$

13.4 Step 3: Extra Current Due to Voltage Difference

$$I_{\text{extra}} = \frac{V_A - V_B}{rL}$$

$$I_{\text{extra}} = \frac{5}{0.0005 \times 300}$$

$$I_{\text{extra}} = \frac{5}{0.15}$$

$$I_{\text{extra}} = 33.33 \text{ A}$$

13.5 Step 4: Current Supplied from A

$$I_A = I_{A0} + I_{\text{extra}}$$

$$I_A = 60 + 33.33$$

$$\boxed{I_A = 93.33 \text{ A}}$$

13.6 Step 5: Current Supplied from B

Total load:

$$I_{\text{total}} = 50 + 80 = 130 \text{ A}$$

$$I_B = 130 - 93.33$$

$$\boxed{I_B = 36.67 \text{ A}}$$

13.7 Step 6: Section Currents

$$I_{AC} = 93.33 \text{ A}$$

$$I_{CD} = 93.33 - 50 = 43.33 \text{ A}$$

$$I_{DB} = 43.33 - 80 = -36.67 \text{ A}$$

Negative sign means actual current in section DB flows from B to D.

13.8 Step 7: Voltage Calculation

Resistance of each 100 m section:

$$R = 100 \times 0.0005 = 0.05\Omega$$

Voltage drop from A to C:

$$\Delta V_{AC} = 93.33 \times 0.05 = 4.67 \text{ V}$$

$$V_C = 250 - 4.67 = 245.33 \text{ V}$$

Voltage drop from C to D:

$$\Delta V_{CD} = 43.33 \times 0.05 = 2.17 \text{ V}$$

$$V_D = 245.33 - 2.17 = 243.16 \text{ V}$$

Check from B side:

$$\Delta V_{BD} = 36.67 \times 0.05 = 1.83 \text{ V}$$

$$V_D = 245 - 1.83 = 243.17 \text{ V}$$

Small difference is due to rounding.

$V_{\min} \approx 243.17 \text{ V at point D}$
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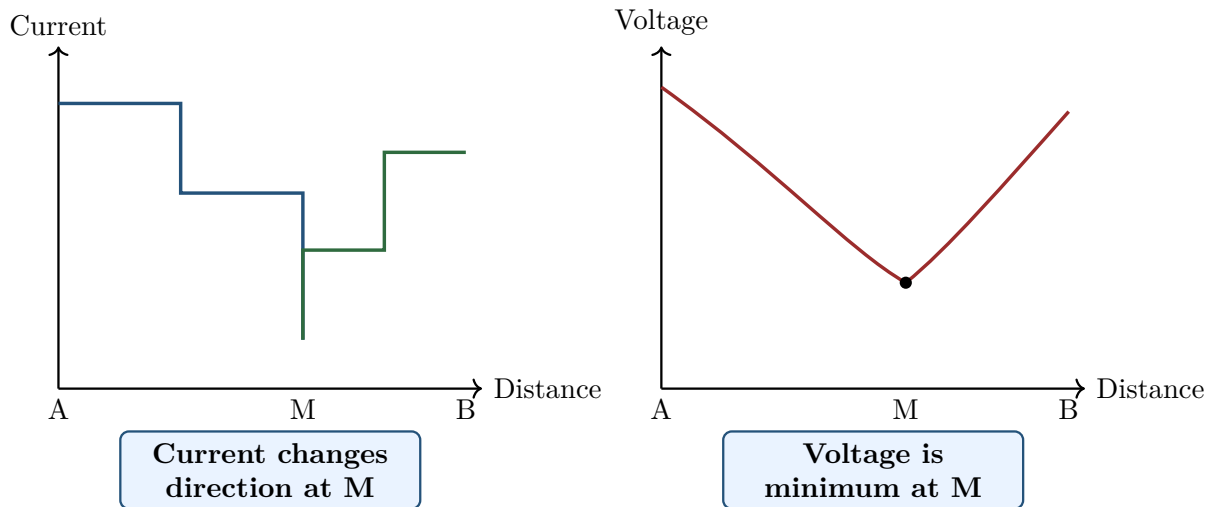
14 Current and Voltage Profile for Both-End Feeding

14.1 Current Profile

In both-end feeding, current decreases from A side and also decreases from B side. At the minimum potential point, current changes direction.

14.2 Voltage Profile

Voltage falls from A towards the minimum point. It also falls from B towards the same minimum point.



How to Read the Diagram

Word/Symbol	Meaning
Current profile	Graph showing how current changes along the distributor.
Voltage profile	Graph showing how voltage changes along the distributor.
Distance	Horizontal axis from A to B.
A	Left feeding end.
B	Right feeding end.
M	Minimum potential point.
Current changes direction at M	Current from A and current from B meet at M.
Voltage is minimum at M	Voltage falls from both ends towards M, so M has the lowest voltage.

15 Case 3: Uniformly Loaded Distributor Fed at Both Ends

Definition

A **uniformly loaded distributor fed at both ends** is a distributor in which load is spread uniformly along the length and supply is given from both ends.

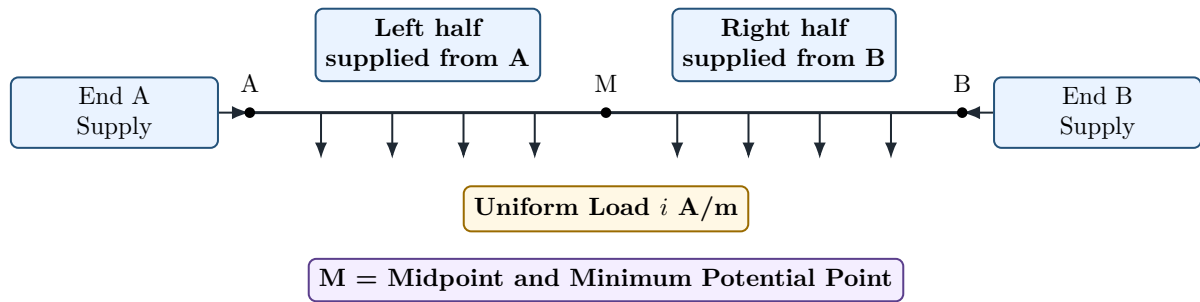
For equal end voltages:

$$V_A = V_B$$

the minimum potential point lies at the middle of the distributor.

Minimum potential point is at midpoint for equal end voltages and uniform load.

16 Uniform Load Diagram



How to Read the Diagram

Word/Symbol	Meaning
End A Supply	Left end supply point.
End B Supply	Right end supply point.
A and B	Two ends of the distributor.
M	Midpoint of AB. For equal end voltages and uniform load, it is the minimum potential point.
Uniform Load i A/m	Every metre length takes i ampere current.
Left half supplied from A	The left half of the uniform load gets current from A.
Right half supplied from B	The right half of the uniform load gets current from B.
Downward arrows	They represent continuous load spread along the distributor.

17 Maximum Voltage Drop for Uniform Load

Let:

l = total length of distributor

i = load current per metre

r = resistance per metre

For equal end voltages, each end supplies half length:

$$\frac{l}{2}$$

Voltage drop from A to midpoint M:

$$\Delta V_{\max} = \frac{ir \left(\frac{l}{2}\right)^2}{2}$$

$$\Delta V_{\max} = \frac{irl^2}{8}$$

If total current is:

$$I = il$$

and total resistance is:

$$R = rl$$

then:

$$\Delta V_{\max} = \frac{IR}{8}$$

Memory Hook

For a uniformly loaded distributor fed at both ends at equal voltage:

$$\text{Maximum voltage drop} = \frac{IR}{8}$$

18 Power Loss for Uniform Load Fed at Both Ends

This result is also important for semester.

For one half of distributor:

$$\text{length} = \frac{l}{2}$$

At a distance x from the midpoint towards one end, current is:

$$ix$$

Small loss:

$$dP = (ix)^2 r dx$$

For both halves:

$$P_{\text{loss}} = 2 \int_0^{l/2} i^2 x^2 r dx$$

$$P_{\text{loss}} = 2i^2 r \left[\frac{x^3}{3} \right]_0^{l/2}$$

$$P_{\text{loss}} = 2i^2 r \frac{l^3}{24}$$

$$P_{\text{loss}} = \frac{i^2 r l^3}{12}$$

Since:

$$I = il, \quad R = rl$$

$$I^2 R = i^2 l^2 \times rl = i^2 r l^3$$

Therefore:

$$P_{\text{loss}} = \frac{I^2 R}{12}$$

Semester Answer Style

This derivation is a very important semester answer. Write the integral cleanly and mention that both halves are identical.

19 Solved Example 3: Uniform Load Fed at Both Ends

19.1 Problem

A 2-wire DC distributor of length 400 m is fed at both ends at 240 V. It has uniformly distributed load of 0.5 A/m. Resistance of distributor is $0.0004\Omega/\text{m}$ for both conductors. Find the minimum voltage.

19.2 Solution

Given:

$$l = 400 \text{ m}$$

$$i = 0.5 \text{ A/m}$$

$$r = 0.0004\Omega/\text{m}$$

Total current:

$$I = il = 0.5 \times 400 = 200 \text{ A}$$

Total resistance:

$$R = rl = 0.0004 \times 400 = 0.16\Omega$$

Maximum voltage drop:

$$\Delta V_{\text{max}} = \frac{IR}{8}$$

$$\Delta V_{\text{max}} = \frac{200 \times 0.16}{8}$$

$$\Delta V_{\text{max}} = 4 \text{ V}$$

Minimum voltage:

$$V_{\min} = 240 - 4$$

$$V_{\min} = 236 \text{ V}$$

The minimum voltage occurs at the midpoint.

20 Uniform Load with Unequal End Voltages

If the two end voltages are unequal:

$$V_A \neq V_B$$

then the minimum potential point does not lie exactly at the midpoint.

Let minimum potential point be at distance x from A.

Then distance from B is:

$$l - x$$

Voltage drop from A to minimum point:

$$\Delta V_A = \frac{irx^2}{2}$$

Voltage drop from B to minimum point:

$$\Delta V_B = \frac{ir(l-x)^2}{2}$$

At the same minimum point:

$$V_A - \frac{irx^2}{2} = V_B - \frac{ir(l-x)^2}{2}$$

Rearranging:

$$V_A - V_B = \frac{ir}{2} [x^2 - (l-x)^2]$$

$$V_A - V_B = irl \left(x - \frac{l}{2} \right)$$

Therefore:

$$x = \frac{l}{2} + \frac{V_A - V_B}{irl}$$

Semester Answer Style

If $V_A > V_B$, then x becomes greater than $l/2$. So the minimum point shifts towards B.

21 General Numerical Method for Both-End Distributor

For both-end distributor numericals, use this sequence:

1. Draw the line diagram.
2. Mark both end voltages.
3. Mark all loads and distances.
4. Decide whether end voltages are equal or unequal.
5. If equal, use moment method.
6. If unequal, add voltage-difference term.
7. Find currents supplied from A and B.
8. Find section currents.
9. Locate the point where current changes direction.
10. Calculate voltage at that point.
11. Box the minimum voltage.

Common Mistake

Do not assume the far end has minimum voltage in both-end feeding. In both-end feeding, the minimum voltage is at the point where current changes direction.

22 Comparison: One-End Feeding and Both-End Feeding

Point	One-End Feeding	Both-End Feeding
Supply point	Supply from one end only	Supply from both ends
Reliability	Lower	Higher
Voltage drop	Higher	Lower
Minimum voltage	Usually at far end	At current reversal point
Current direction	One direction only	From both ends towards minimum point
Calculation	Section current from one side	Current sharing from both ends
Use	Simple and small distribution	Better distribution and ring-main analysis

23 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Distributor fed at both ends	Supply from A and B	Explain with diagram and advantages.
Equal end voltage	$V_A = V_B$	Find current supplied from each end using moment method.

Unequal end voltage	$V_A \neq V_B$	Find current supplied from each end using voltage-difference term.
Minimum potential point	Current reversal point	Locate the point and find minimum voltage.
Uniform loading	Load spread along length	Derive $\Delta V_{\max} = IR/8$.
Power loss	Loss in uniformly loaded distributor	Derive $P_{\text{loss}} = I^2 R/12$.
Voltage profile	Voltage falls from both ends to minimum point	Draw and explain profile.

24 Most Predictable Semester Questions from Phase 5

No.	Question	Marks
1	Explain distributor fed at both ends with diagram.	4
2	What is minimum potential point? Explain with diagram.	3–4
3	A DC distributor is fed at both ends at equal voltage. Find current supplied from both ends and minimum voltage.	8–10
4	A DC distributor is fed at both ends at unequal voltages. Find current supplied from both ends and minimum voltage.	8–10
5	Derive the moment method for distributor fed at both ends.	5
6	For uniformly loaded distributor fed at both ends, derive maximum voltage drop.	5
7	Show that power loss in uniformly loaded distributor fed at both ends is $I^2 R/12$.	5
8	Compare distributor fed at one end and both ends.	4–6
9	Draw current and voltage profile for distributor fed at both ends.	4

25 Exam Answer Templates

25.1 Template 1: Theory Answer

Semester Answer Style

Answer sequence for distributor fed at both ends:

1. Give definition.
2. Draw clean diagram with A and B supplies.
3. Explain loads and sections.
4. Explain current sharing from both ends.
5. Define minimum potential point.
6. Write advantages over one-end feeding.

25.2 Template 2: Equal End Voltage Numerical

Semester Answer Style

Answer sequence:

1. Draw line diagram.
2. Mark loads and distances.
3. Use:

$$I_A = \frac{\sum(\text{load} \times \text{distance from B})}{L}$$

4. Find:

$$I_B = I_{\text{total}} - I_A$$

5. Find section currents.
6. Find current reversal point.
7. Calculate voltage at that point.

25.3 Template 3: Unequal End Voltage Numerical

Semester Answer Style

Answer sequence:

1. Draw line diagram.
2. Mark V_A , V_B , loads and distances.
3. First calculate equal-voltage current from A.
4. Add voltage-difference current:

$$\frac{V_A - V_B}{rL}$$

5. Find I_A , I_B .
6. Find section currents and minimum voltage.

25.4 Template 4: Uniform Loading Derivation

Semester Answer Style

Answer sequence:

1. Draw uniformly loaded distributor fed at both ends.
2. State that minimum point is midpoint for equal end voltages.
3. Treat each half as one-end fed distributor of length $l/2$.
4. Write:

$$\Delta V_{\text{max}} = \frac{ir(l/2)^2}{2}$$

5. Get:

$$\Delta V_{\text{max}} = \frac{irl^2}{8} = \frac{IR}{8}$$

26 Phase 5 Final Memory Sheet

One-Page Memory Sheet

1. Distributor fed at both ends receives supply from A and B.
2. It gives better voltage regulation than one-end feeding.
3. Minimum potential point is where current changes direction.
4. For equal end voltages:

$$I_A = \frac{\sum(\text{load} \times \text{distance from B})}{L}$$

$$I_B = \frac{\sum(\text{load} \times \text{distance from A})}{L}$$

5. For unequal end voltages:

$$I_A = \frac{V_A - V_B}{rL} + \frac{\sum(\text{load} \times \text{distance from B})}{L}$$

6. Uniform load, equal end voltages:

$$\Delta V_{\max} = \frac{IR}{8}$$

7. Power loss for uniformly loaded distributor fed at both ends:

$$P_{\text{loss}} = \frac{I^2 R}{12}$$

8. In both-end feeding, far end is not necessarily minimum voltage point.

27 Phase 5 Self-Test

Answer these without seeing notes:

1. What is a distributor fed at both ends?
2. Why is both-end feeding better than one-end feeding?
3. What is minimum potential point?
4. Why does current change direction at the minimum potential point?
5. Write the moment method formula for equal end voltages.
6. Write the formula for unequal end voltages.
7. How do you locate the minimum voltage point?
8. Derive $\Delta V_{\max} = IR/8$ for uniformly loaded distributor fed at both ends.
9. Derive $P_{\text{loss}} = I^2 R/12$.
10. Compare one-end feeding and both-end feeding.

Common Mistake

In both-end distributor numericals, never start from voltage drop directly. First find current supplied from A and B. Then find section currents. Then calculate voltage.

End of Phase 5

Next Phase:

Phase 6 – Ring Main Distributor

Ring distributor, interconnector, KVL loop method, current distribution and voltage at load points.